

Aggregate Fluctuations from Firm Comovement

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ABSTRACT

This paper demonstrates that correlated idiosyncratic fluctuations among firms are an important source of macroeconomic volatility. Standard methods based on cross-sectional demeaning obscure this channel by mechanically driving average pairwise correlations to zero asymptotically. We develop a nonparametric bounds approach quantifying the contribution of cross-firm comovement within industries directly from firm-level data. Applied to the United States economy, we find that such clustered comovement explains 10–15% of GDP volatility in normal times, 15–30% in downturns, and up to 45% following the Great Recession, while also helping account for the Great Moderation and its undoing.

1. Introduction

One of the central questions in macroeconomics is what drives business-cycle fluctuations. The canonical representative-agent framework attributes aggregate volatility primarily to economy-wide or sector-wide shocks, assuming that firm-specific disturbances wash out through the law of large numbers. Yet recent evidence from firm-level data reveals a much more heterogeneous economy. The firm-size distribution is highly skewed, with a handful of very large firms exerting outsize influence on aggregate outcomes. This observation gave rise to the influential *granular view* (Jovanovic, 1987; Gabaix, 2011), which shows that idiosyncratic shocks to large firms can translate into significant aggregate fluctuations.

In this paper, we uncover another micro-level source of aggregate volatility: within-cluster comovement of firm-specific components, which we term *clustered origins*. This arises when firms experience correlated, though not identical, shocks within groups such as industries. They differ from common macro-shocks, since they do not affect all firms equally. Instead, they reflect a form of pairwise comovement or synchronization within groups. Because these shocks are correlated, they do not diversify. As a result, even moderate-sized firms can collectively generate substantial volatility at the aggregate level.

Why has this channel been underappreciated? A key reason is methodological. Commonly used empirical approaches—most notably cross-sectional demeaning—to isolate firm-specific (idiosyncratic) components mechanically impose an average pairwise correlation across firms toward zero asymptotically. This removes the imprint of clustered comovement by construction, rather than because it is absent. As a result, attenuation bias arises, and the importance of firm-level interdependence is understated. Our first contribution is to formalize and demonstrate this pitfall.

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To address this challenge, we develop a simple and robust nonparametric decomposition that extracts two components of micro origins from aggregate fluctuations: (i) *clustered origins*—the contribution of within-industry comovement of firm-specific shocks, and (ii) *granular origins*—the importance of idiosyncratic shocks to large firms. Our method relies only on basic variance-covariance properties (such as nonnegativity and Cauchy-Schwarz bounds), observable firm- and industry-level moments, and natural economic weights like sales or Domar weights. By construction, it requires no restrictive structural assumptions. The approach delivers robust upper and lower bounds for each source of volatility and is valid under heterogeneity and misspecification.

We apply this framework to U.S. publicly traded firms (Compustat, 1975–2023) and obtain three main results. First, clustered origins are important even during tranquil periods, accounting for roughly 10–15% of U.S. GDP variance. Second, they are countercyclical: their share of aggregate volatility of the U.S. economy rises to 15–30% in downturns and peaks near 45% following turbulent periods such as the Great Recession. Third, their time path closely tracks major shifts in economic volatility: their subsequent decline following the Great Moderation coincides with falling GDP volatility, while their resurgence after 2000 parallels the concurrent rise in aggregate volatility. These results are robust across alternative measures of firm activity (sales, productivity, proxies for value-added), filtering procedures, Domar adjustments, and sectoral subsamples. Although industries differ in the extent of within-industry comovement, no single sector drives the aggregate patterns.

Taken together, our evidence shows that the business cycle is shaped not only by aggregate shocks or the idiosyncratic fortunes of a few large firms, but also by correlated fluctuations across many moderate-sized firms within industries. By documenting the empirical importance and cyclical properties of clustered origins, we provide a new microeconomic foundation for aggregate volatility. This highlights the broader need for macroeconomic analysis to account for firm networks and interdependencies, which can amplify and transmit shocks across the economy.

Contribution to the literature This paper contributes to several key strands of research on business cycle fluctuations by uncovering a novel microeconomic source of aggregate volatility. Our approach builds on recent advances in the heterogeneous-firms literature that relax conventional assumptions about the distribution and dependence of firm-level fluctuations and emphasize complex cross-sectional dependence structures (e.g., Guvenen, Karahan, Ozkan and Song, 2021; Sterk, Sedlacek and Pugsley, 2021; Forneron, 2023; Jaimovich, Terry and Vincent, 2023).

A prominent strand of the business cycle literature isolates firm-level fluctuations by removing cross-sectional common components, implicitly or explicitly treating the resulting residuals as the relevant firm idiosyncratic components. We show that such procedures can mechanically eliminate average residual comovement and therefore obscure a potentially important covariance channel for aggregate volatility. Our contribution is directed specifically at this strand of the literature and the implications of such procedures for measuring the microeconomic origins of aggregate fluctuations.

A substantial body of work emphasizes the role of microeconomic shocks and firm heterogeneity in shaping aggregate volatility. Foundational contributions examine how firm-level shocks and heterogeneity aggregate up to influence macro volatility (e.g., Comin and Philippon, 2005; Comin and Mulani, 2006; Gabaix, 2011; Carvalho and Gabaix, 2013). Within this literature, our paper introduces a complementary microeconomic channel—clustered origins—that enriches the well-established granularity framework. Originating from Jovanovic (1987) and formalized by Gabaix (2011), granularity captures how idiosyncratic shocks to large firms disproportionately affect aggregate fluctuations.¹

Further advancing this line of inquiry, several studies explore the network origins of aggregate fluctuations, incorporating supply chains and input-output linkages that amplify firm-level volatility through interconnections (e.g., Acemoglu, Carvalho, Ozdaglar and Tahbaz-Salehi, 2012; Carvalho, 2014; Oberfield, 2018; Herskovic, Kelly, Lustig and Van Nieuwerburgh, 2020). Additionally, related works examine cross-sector correlations and the hierarchical transmission of shocks across levels of economic aggregation (e.g., Carvalho and Gabaix, 2013; di Giovanni, Levchenko and Mejean, 2014), as well as the micro-level sources of volatility in open economies (e.g., di Giovanni and Levchenko, 2012; Gaubert and Itskhoki, 2021). Complementing these approaches, we focus on within-cluster pairwise comovement among firms and its contribution to aggregate volatility.

Our contribution also extends the literature on the industrial and sectoral origins of macroeconomic volatility, tracing back to Long and Plosser (1983). This research shows that interconnected supply chains and production networks limit the diversification of industry-specific shocks, thereby propagating sectoral disturbances into aggregate

¹For comprehensive treatments of granularity, including applications to financial sectors, see Buch and Neugebauer (2011), Amiti and Weinstein (2018), Bremus, Buch, Russ and Schnitzer (2018), Kim, Park and So (2025), among others.

fluctuations (e.g., Horvath, 1998; Horvath and Verbrugge, 1999; Dopor, 1999; Foerster, Sarte and Watson, 2011; Atalay, 2017). While those studies predominantly focus on between-industry correlations, we complement their insights by uncovering significant micro-level within-industry comovement among firms.

Finally, our empirical findings shed light on the dynamics of U.S. GDP volatility over recent decades. We document a pronounced U-shaped pattern in the volatility—marked by a decline from the early 1980s to the early 2000s, followed by a resurgence thereafter—which we show is closely associated with changes in within-cluster firm comovement. This pattern aligns closely with the literature on the Great Moderation and its aftermath (e.g., Kim and Nelson, 1999; Stock and Watson, 2002; Comin and Mulani, 2006; Davis, Haltiwanger, Jarmin, Miranda, Foote and Nagypal, 2006), consistent with theoretical arguments in Carvalho and Gabaix (2013) that emphasize the amplified role of firm- and industry-level volatility in shaping macroeconomic stability and its reversal.

The remainder of the paper is organized as follows. Section 2 introduces the statistical framework. Section 3 demonstrates the pitfalls of cross-sectional demeaning. Section 4 develops the empirical methodology. Section 5 presents the main empirical findings. Section 6 concludes.

2. Framework and Motivation

This section introduces a simple statistical framework to illustrate how correlated idiosyncratic firm behaviors can generate aggregate fluctuations.

2.1. Framework

Consider a cluster (industry) with N_t number of firms, where firm i 's variable (e.g., sales, output, employment, productivity) is expressed in log form and $\hat{y}_{it}$ denotes its business cycle part of the variable after removing long-run trends. We decompose each firm's fluctuations into two components that are uncorrelated with each other and have zero mean with finite standard deviations $\sigma_{A,t}$ and $\sigma_{F,it}$:

$$\hat{y}_{it} = \varepsilon_{A,t} + \varepsilon_{F,it}, \quad (1)$$

where $\varepsilon_{A,t}$ is a benchmark common component affecting all firms in the cluster. We interpret $\varepsilon_{A,t}$ as the component common to firms within a given industry, including both industry-specific shocks and the industry's response to economy-wide conditions. The other term, $\varepsilon_{F,it}$, is a benchmark firm-specific (idiosyncratic) component. Thus, conditional on belonging to the same cluster, firms are assumed to have the same loading on the cluster-level common component, while their remaining fluctuations $\varepsilon_{F,it}$ may be correlated.

Our key innovation is to relax the conventional assumption that firm-specific components are uncorrelated across firms. We allow the pairwise correlation, $\rho_{FF,ii't} \equiv \text{corr}(\varepsilon_{F,it}, \varepsilon_{F,i't})$, to be nonzero for $i \neq i'$. This correlation can generate aggregate-level fluctuations and foster comovements ($\text{corr}(\hat{y}_{it}, \hat{y}_{i't})$) across firms within the cluster.² This motivates our use of 'cluster' rather than 'industry' terminology to emphasize these within-cluster correlations. Sections 2 and 3 focus on within-cluster fluctuations; Section 4 will extend the analysis to the multi-cluster economy.

Discussions on Our Framework The benchmark framework provides a convenient decomposition for the economic question studied in this paper, but it is not the only available decomposition of firm-level fluctuations. In particular, a factor model can provide an alternative way to represent within-cluster comovement by allowing firms to have heterogeneous exposures to latent common factors. We therefore do not view our framework as ruling out or dominating factor-based decompositions. Rather, the two approaches organize the same observed comovement differently and serve different purposes.

To make the relation precise, consider the modified representation motivated from a factor model with heterogeneous loadings:

$$\hat{y}_{it} = \varepsilon_{A,t} + \underbrace{(\lambda_{it}f_t + u_{it})}_{\equiv \varepsilon_{F,it}}, \quad (2)$$

²These pairwise correlations prevent the average firm-specific component from vanishing, contributing to notable short-term departures of the cross-sectional mean from zero, i.e., $N_t^{-1} \sum_i \varepsilon_{F,it} \neq 0$. However, since correlations affect second moments rather than first moments, they leave the long-run level unaffected: $E[\varepsilon_{F,it}] = 0$ and $\lim_{T \rightarrow \infty} T^{-1} \sum_i \varepsilon_{F,it} = 0$.

where $\varepsilon_{A,t}$ loads equally on all firms, f_t is a latent factor with firm-specific loadings λ_{it} , u_{it} is a firm-specific idiosyncratic component that can be pairwise correlated (i.e., $\text{corr}(u_{it}, u_{i't'}) \neq 0$), and the three are mutually uncorrelated.³

This representation makes clear that heterogeneous exposure to a common factor can be incorporated into the benchmark firm-specific component when the common component is defined as the cluster-level component with a common loading. Thus, the benchmark clustered-origin measure can contain covariance generated by heterogeneous responses to latent common factors, even though it does not separately identify that factor structure. In this sense, $\varepsilon_{F,it}$ is a firm-specific component in our decomposition, but may not be purely idiosyncratic in the strict factor-model sense: it may contain comovement arising from heterogeneous exposure to latent common factors.

A factor decomposition offers a more structured account of the source of comovement, at the cost of additional identifying restrictions: a normalization and assumptions on the number and structure of factors. Our objective is different. We do not aim to identify or classify latent factors; we quantify the contribution of within-cluster pairwise covariance to aggregate volatility. The homogeneous-loading assumption is a simplifying and practical restriction that allows us to address this question without imposing the additional complexity of a fully heterogeneous factor structure. It is not a claim that firms have identical exposure to all aggregate shocks.

The plausibility of this assumption depends on how the cluster is defined. We thus use narrowly defined industries as clusters, where firms face more similar industry-wide conditions than firms across broad sectors or the economy as a whole. We therefore take a bottom-up approach: we first apply the benchmark decomposition separately within each narrowly defined industry and then aggregate the results to the whole economy. Nevertheless, our results under the benchmark specification should be interpreted as conditional on this identifying restriction. If substantial heterogeneity in exposure to common factors remains within an industry, some factor-driven comovement arising from heterogeneous loadings will be captured by $\varepsilon_{F,it}$ and thus attributed to our measure of clustered origins.

2.2. The Aggregate Business Cycle Fluctuations of a Cluster

Cluster-level business cycle volatility arises from three sources: cluster-wide common fluctuations, firm-specific fluctuations, and pairwise comovements of firm-specific fluctuations within a cluster. The cluster-level aggregate business cycle is:

$$\hat{Y}_t = \sum_i w_{it} \hat{y}_{it}, \quad (3)$$

where w_{it} represents the share of firm size within the cluster, and satisfies $\sum_i w_{it} = 1$.⁴ The variance of the cluster-level aggregate business cycle can be decomposed into three terms:

$$\sigma_{\hat{Y},t}^2 = \sigma_{A,t}^2 + \sum_i w_{it}^2 \sigma_{F,it}^2 + \sum_i w_{it} \sum_{i' \neq i} w_{i't} \rho_{FF,ii'} \sigma_{F,ii'} \sigma_{F,i't'}, \quad (4)$$

where the second and third terms represent the micro origins of macro fluctuations through the volatility and comovements of firm-specific fluctuations, respectively. To clarify the economic interpretation, we define three key terms:

Definition 1 (Cluster-wide common origins). $\sigma_{A,t}^2$: The variance of the common component affecting all firms within a cluster.

Definition 2 (Granular origins). $\Gamma_t = \sum_i w_{it}^2 \sigma_{F,it}^2$: The contribution from individual firms' firm-specific components, weighted by squared firm shares (Gabaix, 2011).

Definition 3 (Clustered origins). $\chi_t = \sum_i w_{it} \sum_{i' \neq i} w_{i't} \rho_{FF,ii'} \sigma_{F,ii'} \sigma_{F,i't'}$: The contribution of synchronized idiosyncratic movements to cluster-level aggregate volatility through within-cluster covariance of firm-specific components.

³When $\text{corr}(u_{it}, u_{i't'}) = 0$, comovement among firms' benchmark firm-specific components arises from heterogeneous exposure to the latent factor f_t . This representation is therefore compatible with our framework: factor-generated comovement is one possible source of the pairwise correlation that enters our clustered-origin measure. See Appendix D for more details: under the standard factor-loading normalization, the estimated factor model faces a similar problem as the demeaned process.

⁴In addition to size weights, business cycle research often uses Domar weights to account for input-output linkages (e.g., Domar, 1961; Hulten, 1978). For simplicity, we do not use them here; Section 4 does so.

A Simple Example of Clustered Origins To illustrate the importance of clustered origins, consider a simplified setting with a single cluster in which all firms have identical variances and pairwise covariances of their firm-specific components. This assumption allows us to rewrite equation (4) as:

$$\sigma_{\hat{Y},t}^2 = \sigma_{A,t}^2 + h_t^2 \sigma_{F,t}^2 + (1 - h_t^2) \rho_{F,t} \sigma_{F,t}^2, \quad (5)$$

where $h_t = (\sum_i w_{it}^2)^{1/2}$ represents the Herfindahl-Hirschman index (HHI) within the cluster. This decomposition reveals three distinct sources of aggregate volatility: common components, granular origin from firm size dispersion, and clustered origin from correlated idiosyncratic movements.

The first term of equation (5) captures common origins. The remaining terms are micro origins that represent a convex combination of variance and covariance ($\sigma_{F,t}^2$ and $\rho_{F,t} \sigma_{F,t}^2$) with weight h_t^2 . Specifically, the second term represents granular origins, which, Lucas (1977) argued, should vanish through diversification when $h_t \rightarrow 0$ as $N_t \rightarrow \infty$. However, Gabaix (2011) showed that with the fat-tailed firm size distribution, the HHI remains bounded away from zero even as the number of firms increases, preserving the granular channel.

Our focus is on the third term—clustered origins. Two key insights emerge: First, positive correlations across firms ($\rho_{F,t} > 0$) amplify aggregate volatility. Second, clustered origins and granular origins respond in opposite directions to market concentration: as HHI falls, the granular channel weakens while the clustered channel strengthens. To quantify the relative importance of these channels, consider the ratio of clustered to granular origins:

$$\frac{\chi_t}{\Gamma_t} = \left(\frac{1}{h_t^2} - 1 \right) \rho_{F,t}. \quad (6)$$

For instance, assume a size distribution with an HHI of $h_t = 0.12$, as demonstrated in Gabaix (2011).⁵ In this context, even small positive pairwise correlations of idiosyncratic fluctuations, ranging from 1% to 5%, imply that clustered origins range from 68% to 342% of granular origins, according to equation (6). A 1.46% correlation coefficient results in an equal contribution of firm-specific comovements and granularity to aggregate volatility. These calculations highlight that clustered origins can be one of the key drivers of aggregate business cycle fluctuations, even with seemingly small pairwise correlations.

3. The Problem with Demeaning

In equation (1), a firm's business cycle variable, $\hat{y}_{it}$, is directly observable, but its benchmark common and firm-specific components remain hidden. A standard approach in the business-cycle literature approximates the common component using the cross-sectional mean and defines the firm-specific component as the deviation from that mean. Specifically, we define the pseudo common and firm-specific components as

$$\hat{y}_{it} = (\bar{\hat{y}}_t) + (\hat{y}_{it} - \bar{\hat{y}}_t) \equiv e_{A,t} + e_{F,it}, \quad (7)$$

where $\bar{\hat{y}}_t = N_t^{-1} \sum_i \hat{y}_{it}$ represents the cross-sectional sample mean.

3.1. Simulation Design and Results

We conduct Monte Carlo simulations to illustrate how standard demeaning obscures important dynamics when firm-specific components are correlated. We simulate 5,000 firms over 50 periods, drawing firm-specific components from a multivariate normal distribution with zero mean, 12% standard deviation, and either zero or 2.5% pairwise correlation. Firms have uniform weights, and no common component is included.⁶ We repeat this simulation 3,000 times to obtain robust sample statistics for both benchmark (true) and pseudo-firm-specific components.

Two key findings emerge. First, aggregate fluctuations differ markedly between benchmark (true) and pseudo (demeaned) components. Figure 1 contrasts these patterns across two economies in Panels (a) and (b). While benchmark firm-specific components (orange solid line) exhibit larger aggregate fluctuations when pairwise correlations are positive, pseudo components (blue dashed line) remain at zero by construction in both cases. Second, demeaning masks

⁵In our U.S. public firm-level data, h_t is around 0.085. In that case, the correlation 1% implies that the clustered origins are around 137% of granular origins in equation (6).

⁶For detailed results and additional simulations with varying weights, see Appendix A.

Aggregate Fluctuations from Firm Comovement

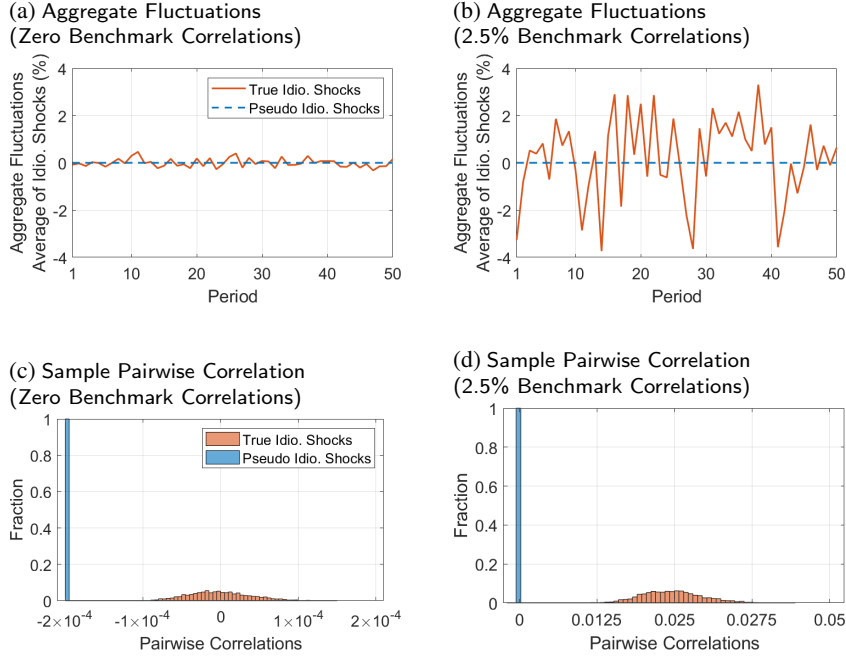


Figure 1: Aggregate Fluctuations and Sample Pairwise Correlations: Benchmark vs. Pseudo Components Notes: Panels (a) and (b) plot the aggregate fluctuations based on the average true and pseudo firm-specific components in each period (orange solid line, $N_t^{-1} \sum_{i=1}^{5,000} \epsilon_{F,it}$, and blue dashed line, $N_t^{-1} \sum_{i=1}^{5,000} e_{F,it}$, respectively). Panels (c) and (d) plot histograms for sample correlations of true and pseudo variables (orange bars, $\text{corr}(\epsilon_{F,it}, \epsilon_{F,i't'})$, and blue bars, $\text{corr}(e_{F,it}, e_{F,i't'})$, respectively) from 3,000 simulations. In Panels (a) and (c), samples are generated from a multivariate normal distribution with zero mean, a 12% standard deviation, and a 2.5% pairwise correlation. In Panels (b) and (d), the pairwise correlation is set to zero.

true correlation patterns. Panels (c) and (d) present the distribution of pairwise correlations from our 3,000 simulations. While the benchmark components retain their imposed correlations (0% and 2.5%), the pseudo components always exhibit near-zero correlations ($\approx -2 \times 10^{-4}$) in both cases, regardless of the underlying correlation structure.⁷ Taken together, these results show that demeaning mechanically eliminates evidence of correlated firm-specific movements, potentially leading researchers to understate an important source of aggregate volatility.

3.2. Non-Negligible Difference Between Benchmark and Pseudo Variables

When benchmark firm-specific components exhibit correlation, pseudo variables derived from demeaning procedures are inadequate proxies for the benchmark variables. We demonstrate this formally by examining the absolute difference between benchmark and pseudo components:

$$|e_{A,t} - \epsilon_{A,t}| = |e_{F,t} - \epsilon_{F,t}| = |\bar{\epsilon}_{F,t}|, \quad (8)$$

where $\bar{\epsilon}_{F,t} = N_t^{-1} \sum_i \epsilon_{F,it}$ represents the cross-sectional average of the benchmark firm-specific components. This equality reveals that non-zero pairwise comovements among firm-specific components generate systematic differences between benchmark and pseudo measures.

To formalize this insight, we introduce the average of idiosyncratic variances defined as $\bar{\sigma}_{F,t}^2 = N_t^{-1} \sum_i \sigma_{F,it}^2$. Also, $\bar{\text{cov}}_{FF,t} = N_t^{-1} \sum_i \bar{\text{cov}}_{FF,it}$ denotes the average covariance of firm-specific components, where $\bar{\text{cov}}_{FF,it} = (N_t - 1)^{-1} \sum_{i' \neq i} \rho_{FF,ii't'} \sigma_{F,it} \sigma_{F,i't'}$ is the average covariance of firm i with all other firms. These definitions allow us to establish the following result:

⁷We discuss the slightly negative correlations of the pseudo components and show that they are asymptotically zero under identical variance-covariance structures ($-1/(N_t - 1)$ in equation 17).

Lemma 1. *The difference between benchmark and pseudo components fails to converge to zero in mean square as the number of firms approaches infinity. For any firm i , the following holds:*

$$\mathbb{E}[|e_{A,t} - \varepsilon_{A,t}|^2] = \mathbb{E}[|e_{F,it} - \varepsilon_{F,it}|^2] = \frac{1}{N_t} \bar{\sigma}_{F,t}^2 + \left(1 - \frac{1}{N_t}\right) \overline{\text{cov}}_{FF,t}. \quad (9)$$

PROOF. The result follows directly from equation (8), with $\mathbb{E}[|\bar{\varepsilon}_{F,t}|^2] = \text{var}(\bar{\varepsilon}_{F,t})$.

This lemma has important implications for business cycle analysis. While the first term on the right-hand side of equation (9) vanishes as the number of firms increases under finite variances, the second term remains non-negligible when average pairwise covariances are non-zero. Consequently, the pseudo variables are inconsistent proxies for the benchmark components whenever $\overline{\text{cov}}_{FF,t} \neq 0$. This renders the conventional approach of using sample means and deviations inadequate for business cycle research.

3.3. Inconsistent Estimator of Variance and Covariance

Building on our previous results, we now demonstrate that pseudo variables yield inconsistent estimators of variance and covariance when comovements exist ($\overline{\text{cov}}_{FF,t} \neq 0$). The pseudo components systematically misrepresent the variance of both common and firm-specific components as follows.

$$\text{var}(e_{A,t}) = \sigma_{A,t}^2 + \frac{1}{N_t} \bar{\sigma}_{F,t}^2 + \left(1 - \frac{1}{N_t}\right) \overline{\text{cov}}_{FF,t} \quad (10)$$

$$\text{var}(e_{F,it}) = \left(1 - \frac{1}{N_t}\right) (\sigma_{F,it}^2 - \overline{\text{cov}}_{FF,it}) - \left[\frac{1}{N_t} (\sigma_{F,it}^2 - \bar{\sigma}_{F,t}^2) + \left(1 - \frac{1}{N_t}\right) (\overline{\text{cov}}_{FF,it} - \overline{\text{cov}}_{FF,t})\right] \quad (11)$$

These equations reveal two important patterns. First, when firms show no correlation with others ($\overline{\text{cov}}_{FF,it} = 0$), pseudo variables provide consistent estimates. However, if firm-level idiosyncratic components are positively correlated, the pseudo (demeaned) variables used in traditional analyses will systematically overestimate common component volatility while underestimating firm-specific volatility. This mismeasurement can mask important sources of business cycle fluctuations.

More fundamentally, pseudo variables also misrepresent comovements. The covariance between the pseudo common component and firm i 's idiosyncratic component is:

$$\text{cov}(e_{F,it}, e_{A,t}) = \frac{1}{N_t} (\sigma_{F,it}^2 - \bar{\sigma}_{F,t}^2) + \left(1 - \frac{1}{N_t}\right) (\overline{\text{cov}}_{FF,it} - \overline{\text{cov}}_{FF,t}). \quad (12)$$

While this averages to zero ($N_t^{-1} \sum_i \text{cov}(e_{F,it}, e_{A,t}) = 0$), individual firms can show nonzero covariance when their covariances differ from the average ($\overline{\text{cov}}_{FF,it} \neq \overline{\text{cov}}_{FF,t}$), even though benchmark common and firm-specific components are uncorrelated by definition.

The covariance between the pseudo firm-specific components of firm i and i' is:

$$\begin{aligned} \text{cov}(e_{F,it}, e_{F,i't}) &= \rho_{FF,ii't} \sigma_{F,it} \sigma_{F,i't} - \frac{1}{2} \left(1 - \frac{1}{N_t}\right) (\overline{\text{cov}}_{FF,it} + \overline{\text{cov}}_{FF,i't}) \\ &\quad - \frac{1}{2N_t} (\sigma_{F,it}^2 + \sigma_{F,i't}^2) - \frac{1}{2} \text{cov}(e_{F,it} + e_{F,i't}, e_{A,t}). \end{aligned} \quad (13)$$

This leads to our key result:

Proposition 1. *The cross-sectional average of pairwise covariances of pseudo firm-specific components (cross-sectionally demeaned fluctuations) converges to zero as the number of firms goes to infinity.*

PROOF. Combining equation (12) into equation (13) yields:

$$\frac{1}{N_t} \sum_i \frac{1}{N_t - 1} \sum_{i' \neq i} \text{cov}(e_{F,it}, e_{F,i't}) = \frac{1}{N_t} (\overline{\text{cov}}_{FF,t} - \bar{\sigma}_{F,t}^2) \rightarrow 0 \quad \text{as } N_t \rightarrow \infty. \quad (14)$$

This result shows that empirical methods relying on demeaned firm fluctuations mechanically erases evidence of within-cluster comovement. Researchers relying on such methods may falsely conclude that firm-specific components are uncorrelated, thereby overlooking a meaningful source of aggregate fluctuations.

3.4. When Can Demeaning Be Justified? Identical Variance-Covariance

Most existing business cycle studies assume zero or negligible cross-firm correlations. We show that cross-sectional demeaning is defensible only under a restrictive special case: when benchmark firm-specific components have identical variance and covariance across firms.

Suppose all firms have identical variance and covariance for their firm-specific components, i.e., $\sigma_{F,it}^2 = \sigma_{F,t}^2$ and $\rho_{FF,ii't} = \rho_{F,t}$ for all $i \neq i'$. Under these conditions, equations (10) and (11) simplify to:

$$\text{var}(e_{A,t}) = \sigma_{A,t}^2 + \frac{1}{N_t} \sigma_{F,t}^2 + \left(1 - \frac{1}{N_t}\right) \rho_{F,t} \sigma_{F,t}^2, \quad (15)$$

$$\text{var}(e_{F,it}) = \left(1 - \frac{1}{N_t}\right) (1 - \rho_{F,t}) \sigma_{F,t}^2. \quad (16)$$

While pseudo variables still overstate common component volatility and understate firm-specific component volatility when true correlation of benchmark variables is positive, they maintain two useful properties. First, pseudo common and firm-specific components are uncorrelated. Second, cross-firm correlation of pseudo firm-specific components becomes negligible as the number of firms increases.

Specifically, the correlation structure simplifies as follows.⁸

$$\text{corr}(e_{F,it}, e_{A,t}) = 0 \quad \text{and} \quad \text{corr}(e_{F,it}, e_{F,i't}) = -\frac{1}{N_t - 1} \quad (17)$$

These correlations are independent of idiosyncratic correlations of benchmark firm-specific components, and the cross-firm correlation of pseudo firm-specific components converges to zero as the number of firms increases.

Under identical volatility and covariance structures, cross-sectional demeaning reasonably separates common and firm-specific components, yielding the following the cluster's volatility decomposition:

$$\sigma_{\hat{Y},t}^2 = \text{var}(e_{A,t}) + h_t^2 \text{var}(e_{F,it}) - \frac{1 - h_t^2}{N_t - 1} \text{var}(e_{F,it}). \quad (18)$$

However, this assumption rarely holds in practice. Extensive empirical evidence documents substantial heterogeneity in firm-level second moments of key economic variables, including productivity, output, and financial performance (e.g., Stanley, Amaral, Buldyrev, Havlin, Leschhorn, Maass, Salinger and Stanley, 1996; Xu and Malkiel, 2003; Comin and Philippon, 2005; Comin and Mulani, 2006; Chun, Kim, Morck and Yeung, 2008; Castro, Clementi and Lee, 2015; Tweedle, 2018; Sterk et al., 2021). In the presence of heterogeneous variance and covariance structures, cross-sectional demeaning does not accurately recover the common and firm-specific components, necessitating the alternative approach developed in the following section.

4. Identifying Micro Origins of Macro Fluctuations

4.1. Within-Cluster Micro Origins

Within a cluster, the observed firm-level fluctuations ($\{\hat{y}_{it}\}_{i=1}^{N_t}$) provide two key moments:

$$\text{var}(\hat{y}_{it}) = \sigma_{A,t}^2 + \sigma_{F,it}^2 \quad (19)$$

$$\text{cov}(\hat{y}_{it}, \hat{y}_{i't}) = \sigma_{A,t}^2 + \rho_{FF,ii't} \sigma_{F,it} \sigma_{F,i't} \quad (20)$$

⁸Note that the results in equation (17) hold when the pseudo factors are constructed from the unweighted mean. If the weights are unequal, i.e., with a weighted mean, then the pseudo common and firm-specific components are correlated, while the pairwise correlation of pseudo firm-specific components is independent of the benchmark firm-specific components' pairwise correlation. See Appendix B for the related results with the weighted mean.

Using them, we can reformulate the clustered and granular origins from equation (4):

$$\chi_t = \sum_i w_{it} \sum_{i' \neq i} w_{i't} \text{cov}(\hat{y}_{it}, \hat{y}_{i't}) - (1 - h_t^2) \sigma_{A,t}^2 \quad (21)$$

$$\Gamma_t = \sum_i w_{it}^2 \text{var}(\hat{y}_{it}) - h_t^2 \sigma_{A,t}^2 \quad (22)$$

While all terms except $\sigma_{A,t}^2$ are observable, we can bound this parameter:

Proposition 2. *In a cluster, the variance of the common component does not exceed $\tilde{\sigma}_{A,t}^2$.*

$$0 \leq \sigma_{A,t}^2 \leq \tilde{\sigma}_{A,t}^2 = \min_{i,i'} \{ \text{var}(\hat{y}_{it}), [1 + \text{corr}(\hat{y}_{it}, \hat{y}_{i't})] \text{sd}(\hat{y}_{it}) \text{sd}(\hat{y}_{i't}) \} \quad (23)$$

PROOF. First, non-negative variance implies $\text{var}(\hat{y}_{it}) \geq \sigma_{A,t}^2$ in equation (19). Thus, we obtain $\min_i \{ \text{var}(\hat{y}_{it}) \} \geq \sigma_{A,t}^2$. Second, $\rho_{FF,ii't} \in [-1, 1]$ yields $\rho_{FF,ii't} \sigma_{F,ii't} \sigma_{F,i't} \geq -\sigma_{F,ii't} \sigma_{F,i't}$. From equation (20), we obtain $\text{cov}(\hat{y}_{it}, \hat{y}_{i't}) + \sigma_{F,ii't} \sigma_{F,i't} \geq \sigma_{A,t}^2$. Because non-negative $\sigma_{A,t}^2$ implies $\text{var}(\hat{y}_{it}) \geq \sigma_{F,ii't}^2$ for all i , we obtain that $[1 + \text{corr}(\hat{y}_{it}, \hat{y}_{i't})] \text{sd}(\hat{y}_{it}) \text{sd}(\hat{y}_{i't}) \geq \sigma_{A,t}^2$ for any i and i' . Thus, $\min_{i,i'} \{ [1 + \text{corr}(\hat{y}_{it}, \hat{y}_{i't})] \text{sd}(\hat{y}_{it}) \text{sd}(\hat{y}_{i't}) \} \geq \sigma_{A,t}^2$. Hence, we obtain equation (23).

Using the range of $\sigma_{A,t}^2$, we can derive ranges for the clustered and granular origins.

Corollary 1. *The clustered and granular origins are bounded as follows:*

$$\sum_i w_{it} \sum_{i' \neq i} w_{i't} \text{cov}(\hat{y}_{it}, \hat{y}_{i't}) - (1 - h_t^2) \tilde{\sigma}_{A,t}^2 \leq \chi_t \leq \sum_i w_{it} \sum_{i' \neq i} w_{i't} \text{cov}(\hat{y}_{it}, \hat{y}_{i't}) \quad (24)$$

$$\sum_i w_{it}^2 \text{var}(\hat{y}_{it}) - h_t^2 \tilde{\sigma}_{A,t}^2 \leq \Gamma_t \leq \sum_i w_{it}^2 \text{var}(\hat{y}_{it}) \quad (25)$$

PROOF. This is directly from equations (21) and (22) with Proposition 2.

4.2. Aggregation: The Whole Economy's Micro Origins

To measure micro origins in aggregate GDP fluctuations, we aggregate across clusters as follows. Let $i \in I_{st} \subset I_t = \cup_{s' \in S} I_{s't}$ index firms, where $s \in S$ indexes clusters. Define $w_{it} = w_{st} w_{sit}$ as firm i 's economy-wide share, decomposed into its within-cluster share (w_{sit}) and its cluster's share (w_{st}). Then, the whole economy's business cycle, denoted by $\widehat{\text{GDP}}_t$, is computed as $\widehat{\text{GDP}}_t = d_t \sum_{i \in I_t} w_{it} \hat{y}_{it}$ and equivalently expressed as $\widehat{\text{GDP}}_t = d_t \sum_{s \in S} w_{st} \hat{Y}_{st}$, where $\hat{Y}_{st}$ is the cluster-level aggregate business cycle and d_t is the Domar weight adjustment defined as the ratio of gross output to aggregate value-added (GDP), which is widely used in business cycle research to account for both direct and indirect supply chain impacts (e.g., Domar, 1961; Hulten, 1978).

The variance decomposition of aggregate GDP fluctuations is

$$\text{var}(\widehat{\text{GDP}}_t) = d_t^2 \sum_{s \in S} w_{st}^2 \text{var}(\hat{Y}_{st}) + d_t^2 \sum_{s \in S} w_{st} \sum_{s' \in S \setminus \{s\}} w_{s't} \text{cov}(\hat{Y}_{st}, \hat{Y}_{s't}) \quad (26)$$

$$= d_t^2 \sum_{s \in S} w_{st}^2 [\sigma_{A,st}^2 + \chi_{st} + \Gamma_{st}] + \text{BIO}_t, \quad (27)$$

where BIO_t denotes the between-industry origins. The granular and clustered origins of cluster s are $\Gamma_{st} = \sum_{i \in I_{st}} w_{sit}^2 \sigma_{F,ii't}^2$ and $\chi_{st} = \sum_{i \in I_{st}} w_{sit} \sum_{i' \in I_{st} \setminus \{i\}} w_{s'i't} \rho_{FF,ii't} \sigma_{F,ii't} \sigma_{F,i't}$.

Finally, we compute the micro origins of the whole economy as follows:

Definition 4 (Granular origins in the whole economy). $\Gamma_t = d_t^2 \sum_{s \in S} w_{st}^2 \Gamma_{st}$.

Definition 5 (Clustered origins in the whole economy). $\chi_t = d_t^2 \sum_{s \in S} w_{st}^2 \chi_{st}$.

Definition 6 (Aggregate cluster-level common origins across clusters). $d_t^2 \sum_{s \in S} w_{st}^2 \sigma_{A,st}^2$.

The between-industry origins are defined as:

$$\text{BIO}_t = d_t^2 \sum_{s \in S} w_{st} \sum_{s' \in S \setminus \{s\}} w_{s't} \left[\text{cov}(\varepsilon_{A,st}, \varepsilon_{A,s't}) + \sum_{i \in I_{st}} w_{sit} \sum_{i' \in I_{s't}} w_{si't} \text{cov}(\varepsilon_{F,it}, \varepsilon_{F,i't}) \right]. \quad (28)$$

This term BIO_t includes both correlated industry-specific fluctuations across industries (clusters) and cross-cluster firm correlation. While the first component has been widely studied, this paper primarily focuses on the clustered origins related to cross-firm correlation within a cluster, leaving the issues related to BIO_t for future research.

5. The Evolution of Clustered Origins in the U.S. Economy

5.1. Data and Summary Statistics

Our analysis uses annual firm-level sales and employment data from the Compustat North America Fundamental Annuals database for the period 1975–2023. Because the COVID-19 pandemic caused significant disruptions and structural changes, our baseline analysis focuses on 1975–2019. We convert nominal values to real terms using industry-specific price deflators from the U.S. Bureau of Economic Analysis (BEA). We group firms into 53 industry clusters, with grouping determined by BEA deflator availability. Table A2 lists these clusters, and Appendix C provides details on the data construction and measurement methodology.

The Business Cycle and Volatility Measurements We estimate the following panel regression equation, where the residual, $\hat{y}_{it}$, represents the cyclical component.

$$y_{it} = \beta_s y_{it-1} + \psi_s^{\text{age}} \times \ln(t - \text{first year}_i) + \psi_s^{\text{emp}} \times \ln \text{emp}_{it} + \psi_s^{\text{time}} \times t + \delta_i + \hat{y}_{it}, \quad (29)$$

where y_{it} is the logarithm of real sales or labor productivity (real sales per employee) for firm i at time (year) t . The firm fixed effect, δ_i , absorbs time-invariant firm characteristics such as location and cohort, but the regression excludes time and industry-time fixed effects.⁹ Following Castro et al. (2015), we control for firm age ($t - \text{first year}_i$) and size (log employment, $\ln \text{emp}_{it}$), as prior literature documents their negative correlation with firm growth and volatility.¹⁰ All coefficients are estimated separately for each cluster s , allowing for heterogeneity in firm dynamics across industries.

From these cyclical components, we compute firm-specific variances, pairwise covariances, and correlations using an 11-year rolling window centered at each year $t \pm 5$ (i.e., over the interval $[t - 5, t + 5]$). We denote these time-varying moments as $\text{var}(\hat{y}_{it})$, $\text{cov}(\hat{y}_{it}, \hat{y}_{i't})$, $\text{corr}(\hat{y}_{it}, \hat{y}_{i't})$. As a robustness check, we confirm our results using simple growth rates (i.e., log difference, $\hat{y}_{it} = y_{it} - y_{it-1}$) instead of the regression residuals from equation (29). This alternative choice does not materially affect our main conclusions.

Summary Statistics Table 1 reveals two notable patterns. First, average firm-level volatility increased during the Great Moderation period (1986–2000) compared to the early 1980s, aligning with findings by Comin and Philippon (2005) and Comin and Mulani (2006), despite a concurrent decline in volatility of the U.S. business cycle fluctuations. Second, average within-cluster correlation follows a U-shaped pattern: declining from the early 1980s through the 1990s, then doubling after 2000. However, these raw moments conflate firm-specific and common cluster-level fluctuations (equations 19 and 20), so they do not isolate the synchronized component of firm behavior. The following analysis decomposes these moments to separately identify the contribution of within-cluster comovement to aggregate volatility.

To assess the homogeneous variance-covariance assumption discussed in Section 3, we examine the cross-sectional distribution of firm-level moments. If idiosyncratic fluctuations were characterized by a common variance-covariance structure within each cluster, we would expect relatively little dispersion in the observed second moments, $\text{std}(\hat{y}_{it})$ and $\text{corr}(\hat{y}_{it}, \hat{y}_{i't})$. Figure 2 provides strong evidence against this implication by plotting the distributions of firm-level

⁹It is worth noting that our regression with fixed effects does not imply the cross-sectional demeaning process criticized in Section 2. Proposition 1 shows that time or industry-time fixed effects (δ_i or δ_{st}) should be excluded, as they mechanically distort pairwise comovement. However, the firm fixed effect implies that an individual firm's mean business-cycle component is zero over the sample period; that is, its long-run expectation equals zero. This does not imply that the cross-sectional mean of the cyclical component is zero in any given period.

¹⁰For evidence on U.S. manufacturing, see Evans (1987) and Hall (1987). These controls, in conjunction with fixed effects and a linear time trend, also serve to mitigate potential biases from endogenous comovement between aggregate and firm-level productivity driven by mechanisms like technology adoption or vintage capital effects (e.g., Mullen, 2020; Fiori and Scoccianti, 2021).

Table 1
Summary Statistics

Variable	Full sample	Subperiod sample			Additional
	1980–2014	1980–1985	1986–2000	2001–2014	2015–2018
Panel A. Within-firm volatility: standard deviation, $\text{std}(\hat{y}_{it})$					
Mean	0.184	0.160	0.187	0.190	0.181
Standard deviation	0.208	0.153	0.212	0.222	0.222
Quantile 10%	0.056	0.055	0.054	0.058	0.051
50%	0.125	0.119	0.128	0.125	0.117
90%	0.344	0.284	0.352	0.361	0.345
Observations (firms)	78,384	12,814	32,474	33,096	8,340
Panel B. Within-cluster comovement: pairwise correlation: $\text{corr}(\hat{y}_{it}, \hat{y}_{it'})$					
Mean	0.139	0.124	0.092	0.184	0.161
Standard deviation	0.354	0.354	0.341	0.350	0.360
Quantile 10%	−0.352	−0.352	−0.371	−0.304	−0.341
50%	0.152	0.128	0.101	0.207	0.184
90%	0.599	0.596	0.544	0.636	0.629
Observations (pairs)	8,256,384	1,086,406	3,093,306	4,076,672	956,670

Notes: The table reports summary statistics for firm-level volatility (standard deviation of $\hat{y}_{it}$) and within-cluster pairwise comovement (correlation of $\hat{y}_{it}$ and $\hat{y}_{it'}$). Cyclical components $\hat{y}_{it}$ are residuals from the regression of log real sales in equation (29). All moments are computed over an 11-year rolling window $([t-5, t+5])$. In Panel A, the standard deviations are winsorized at the top 1%. In Panel B, pairwise correlations are calculated only for firms within the same cluster (among 53 clusters).

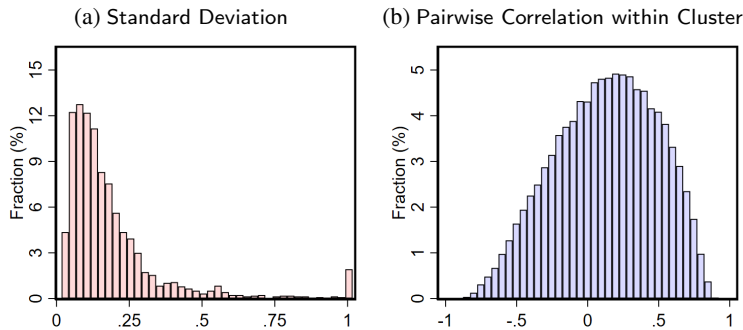


Figure 2: Histograms: Firm Volatility and Comovements Notes: The figures plot histograms of firm volatility (Panel a) and pairwise correlations (Panel b) for 1995. Moments are first calculated for each firm (standard deviation, winsorized by one) and each intra-cluster firm pair (correlation) using data from the 1990–2000 window.

standard deviations and pairwise correlations in 1995.¹¹ Both distributions exhibit substantial dispersion, indicating pronounced heterogeneity in firm-level volatility and comovement. Similar patterns emerge throughout the sample period.

5.2. Main Results: Clustered Origins of the U.S. Business Cycle

We quantify the contribution of clustered origins to aggregate volatility of the whole (private) economy in the U.S. using the framework developed in Section 4. The results point to a quantitatively important and previously underappreciated source of macroeconomic fluctuations: synchronized idiosyncratic movements among firms within

¹¹The same conclusion holds when moments are computed using within-cluster demeaned fluctuations; see Figure A4.

Table 2
The Contribution of Firm Comovement to U.S. Business Cycles

Variable (Sample average)	Full sample	Subperiod sample			Additional
	1980–2014	1980–1985	1986–2000	2001–2014	2015–2018
Panel A. GDP volatility (std. dev., %): $\text{std}(\widehat{\text{GDP}}_t)$					
AR(2) residuals	1.78	2.66	1.53	1.68	2.02
AR(1) residuals	1.87	2.74	1.59	1.80	1.86
Growth rates	1.94	2.83	1.66	1.89	1.94
Panel B. Aggregate cluster-wide common origins (std. dev., %): $d_t(\sum_{s \in S} w_{st}^2 \hat{\sigma}_{A,st}^2)^{1/2}$					
Upper-bound	0.53 (0.039)	0.88 (0.078)	0.43 (0.018)	0.49 (0.022)	0.45 (0.031)
Panel C. Clustered origins (std. dev., %): $\sqrt{ \chi_t } \times \text{sign}(\chi_t)$					
Upper-bound	0.81	1.24	0.60	0.85	0.84
Lower-bound	0.63 (0.050)	0.91 (0.094)	0.44 (0.024)	0.72 (0.031)	0.73 (0.056)
Panel D. Ratio of clustered origin variance to GDP variance (%): $\chi_t / \text{var}(\widehat{\text{GDP}}_t)$					
Upper-bound	21.7	22.0	16.4	27.7	22.2
Lower-bound	13.8 (1.98)	11.9 (2.90)	8.6 (1.17)	20.2 (1.86)	14.3 (2.96)

Notes: This table reports the contribution of within-cluster firm comovement (the ‘clustered origin,’ χ_t) to aggregate U.S. business cycle volatility. In Panel A, GDP volatility is the standard deviation of the cyclical component of logarithmic private GDP, estimated via an autoregressive regression with a linear time trend. In Panel B, aggregate cluster-wide common origin upper bounds (std. dev.) are calculated as the contribution of the whole economy and industry common components in equation (27). Bootstrap standard errors for the sample period averages are reported in parentheses. In Panel C, clustered origins are reported in terms of standard deviations. In Panel D, the ratio measures the contributions of clustered origins as a percentage of total GDP variance. In Panels C and D, the standard errors of lower bounds in parentheses are based on the double bootstrap procedure for aggregate cluster-wide common origin upper-bounds.

the same industry. The estimated upper and lower bounds imply that within-cluster firm comovement accounts for roughly 10–20% of U.S. private GDP volatility over the 1980–2014 sample period (Table 2).

GDP Volatility and Aggregate Cluster-Wide Common Origins Across Clusters Panel A of Table 2 documents the well-known U-shaped pattern of GDP volatility: roughly 2.7% in the early 1980s, declining sharply to approximately 1.5% during the Great Moderation (1986–2000), and rising again through the 2000s and 2010s.¹² Panel B shows that aggregate cluster-wide common origins—the whole economy and industry-level common component volatilities—track only part of this pattern. They declined substantially from 0.88% to 0.43% between 1980–1985 and 1986–2000, consistent with the Great Moderation. However, aggregate cluster-wide common origins remained relatively stable after 2000 and do not account for the subsequent rebound in aggregate volatility, pointing to microeconomic sources as an increasingly important driver of business-cycle fluctuations.

Subperiod Evolution of Clustered Origins and the Great Moderation Clustered origins closely track the U-shaped pattern of the US GDP volatility documented in Panel A of Table 2. During the Great Moderation (1986–2000), the upper- and lower-bound range estimate averaged 0.44–0.60% (in standard deviation units), substantially below the bounds observed in both the preceding and subsequent periods. Over the full sample, clustered origins account for

¹²Aggregate GDP volatility is measured as the standard deviation of the U.S. private economy’s logarithmic GDP business-cycle component, computed using a rolling window of ± 5 years. The business-cycle component is obtained from the residuals of an AR(2) process with a linear time trend. See Figure A5 for the business-cycle components and alternative measures of aggregate GDP volatility from band- or high-pass filters.

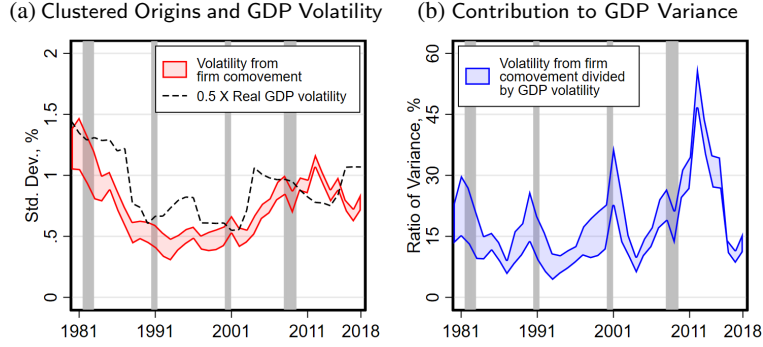


Figure 3: The Evolution of Clustered Origins Notes: The time-series figure displays the contribution of within-cluster firm comovement (the “clustered origin,” χ_t) to aggregate U.S. business-cycle volatility. Panel (a) plots the contribution from clustered origins (red area). The clustered origin’s contribution to the standard deviation is calculated as $100 \times \sqrt{|\chi_t|} \times \text{sign}(\chi_t)$. To compare it with the evolution of aggregate volatility of the whole economy, the black dashed line plots one-half of the standard deviation of aggregate real GDP growth, $0.5 \times 100 \times \text{std}(\widehat{\text{GDP}}_t)$. GDP volatility is measured as the 11-year rolling standard deviation of the cyclical component of private GDP, estimated using an AR(2) process. Panel (b) plots the clustered origin’s contribution as a percentage of GDP variance, $100 \times \chi_t / \text{var}(\widehat{\text{GDP}}_t)$. Shaded gray areas denote NBER recession dates.

13.4–21.7% of GDP variance, falling to 8.6–16.4% during the Great Moderation (Panels C and D).¹³ These parallel movements indicate that shifts in aggregate volatility of the U.S. economy reflect, in part, changes in within-industry firm synchronization over the past four decades.

Countercyclicality of Clustered Origins and the Great Recession The importance of clustered origins becomes particularly pronounced during periods of economic stress. In Figure 3, the contribution of clustered origins to GDP variance (Panel b) reveals a clear countercyclical pattern. During normal periods, clustered origins typically account for roughly 10–15% of aggregate variance. During NBER-dated recessions, however, their contribution frequently rises to 15–30%, suggesting that firm-level comovement becomes more important during downturns. Most notably, following the Great Recession, clustered origins accounted for more than 45% of GDP variance, indicating a high degree of synchronization among firms within industries. These findings highlight a potentially important microeconomic channel through which aggregate volatility is amplified during periods of economic turbulence.

Statistical Issues: Bootstrapping While our nonparametric approach is tractable, it relies on extreme-order statistics that can be sensitive to finite-sample noise. As defined in equations (23) and (24), the upper bound of the aggregate cluster-wide common origins and lower bound of the clustered origins depend on the minimum variance-covariance values within a cluster. Given our 11-year rolling windows, clusters often contain small samples in which random sampling variability can produce firms with very low realized volatility. Because the bound is tied to the minimum estimated variance in the cluster, these low-variance realizations mechanically depress the estimated common component, even if the true underlying comovement is substantial.

To address this concern, we implement a double-resampling bootstrap procedure that combines cluster and block resampling within each cluster for every window.¹⁴ This procedure is valid in panel-data settings with cross-sectional and/or temporal heterogeneity, including environments with spatial dependence (Hounkannounon, 2008; Kapetanios, 2008). Using the bootstrap samples, we construct confidence intervals for cluster-level common-component variance bounds, as well as for the aggregate macro-origin upper bound and clustered-origin lower bound. Figure 4 shows that the nonlinear nature of the bounds induces some skewness in the bootstrap distributions. Nevertheless, the results

¹³Motivated by the GDP volatility decomposition in equation (27), we report contributions in variance terms, i.e., $100 \times \chi_t / \text{var}(\widehat{\text{GDP}}_t)$.

¹⁴We first perform cross-sectional resampling (also referred to as cluster resampling), followed by block resampling of the time series within each cluster for every window. Based on the bootstrap samples, we construct each cluster’s share, w_{st}^* , the corresponding Domar adjustment, d_t^* , and the upper bound on common-factor variance, $\tilde{\sigma}_{A,st}^{*2}$. We then construct the lower bound of clustered origins, $\sum_{i \in I_{st}} w_{sit} \sum_{i' \neq i \in I_{st}} w_{sit'} \text{cov}(\hat{y}_{it}, \hat{y}_{i't}) - (1 - h_t^{*2}) \tilde{\sigma}_{A,st}^{*2}$. Aggregating these components using the bootstrapped weights (w_{st}^* , d_t^*), we finally calculate bootstrap standard errors and confidence intervals.

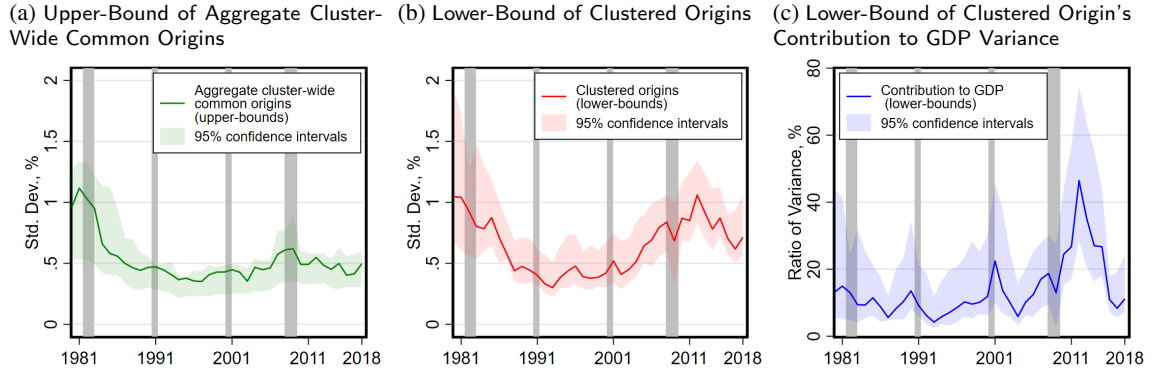


Figure 4: The Aggregate Cluster-Wide Common and Clustered Origins with Confidence Intervals Notes: The figure displays confidence intervals around the aggregate cluster-wide common origin upper bound, the corresponding clustered origin and its contribution (a percentage of GDP variance) lower bounds. Panels (a), (b), and (c) plot these bounds alongside 95% bootstrapped confidence intervals. Shaded gray areas denote NBER recession dates.

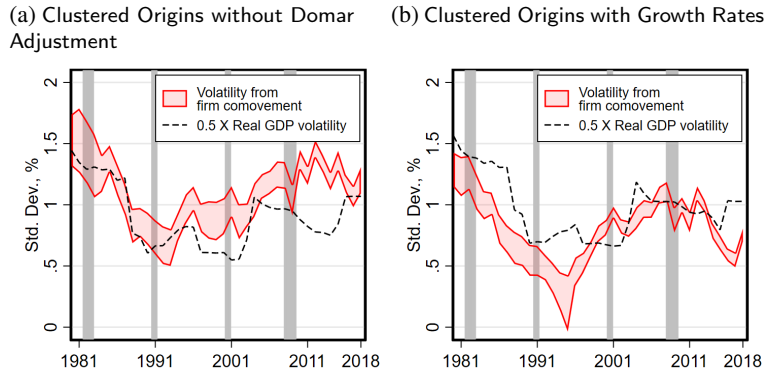


Figure 5: Robustness Check Notes: Panel (a) recalculates the clustered origin contribution without applying the Domar adjustment, i.e., $d_i = 1$ in definition 5. Panel (b) recalculates the main result using simple log-differences (growth rates) for both firm-level and aggregate GDP data instead of AR(2) regression residuals. Shaded gray areas denote NBER recession dates.

confirm that the estimated contribution of clustered origins in Table 2 is statistically significant and that its time-series variation is robust to sampling uncertainty.

5.3. Robustness Checks

Decomposing the Clustered Origin We examine whether the results are driven by two key components of equation (27): firm-level comovement and the Domar adjustment reflecting the economy's input-output structure. Panel (a) of Figure 5 shows that removing the time-varying Domar adjustment leaves the temporal pattern intact—declining during the Great Moderation and rebounding thereafter—indicating that the adjustment affects absolute levels but not the underlying dynamics.¹⁵ Market concentration (measured by HHI) shows only modest variation of 7.5–9% over the sample period (Figure A3's Panel a), suggesting that changes in industry structure are unlikely to explain the observed shifts in clustered origins.

Using Growth Rates We verify the robustness of our findings using simple growth rates instead of regression residuals. Panel (b) of Figure 5 shows that firm-level growth rates (log-differences) produce patterns similar to those

¹⁵The Domar adjustment varies between 50–80% over time, exhibiting a U-shaped pattern consistent with the evolution of clustered origins; see Figure A3's Panel (b).

Aggregate Fluctuations from Firm Comovement

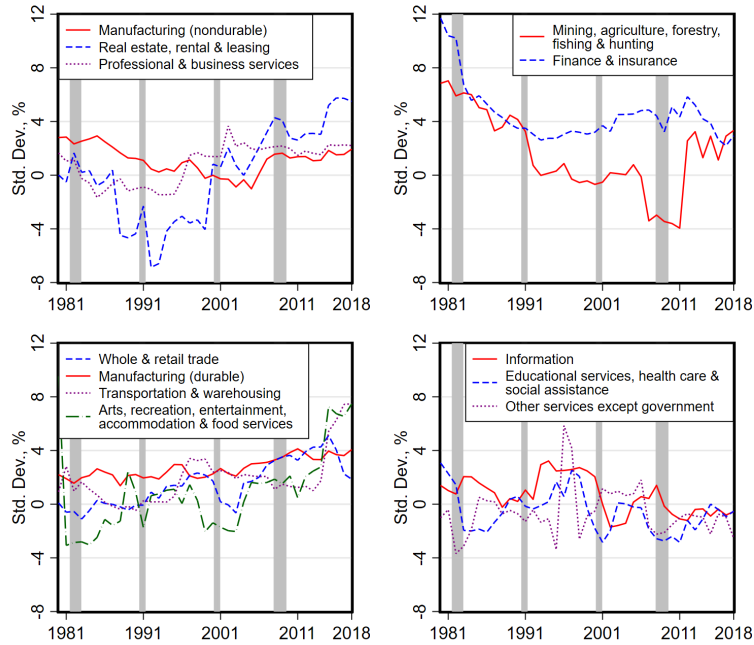


Figure 6: The Evolution of Clustered Origins by Sector Notes: Each panel plots sector's (broadly defined 12 industry groups in Table A1) contribution of firm comovement within cluster to aggregate volatility, measured by the percent standard deviation. For clearer visualization, we plot the midpoint between the upper and lower bounds of clustered origins. Shaded areas denote NBER recession dates.

obtained under our baseline specification. In particular, the close association between clustered origins and aggregate GDP volatility persists under this alternative measure. The result indicates that clustered origins are a robust feature of U.S. business cycle dynamics rather than an artifact of a particular filtering or measurement procedure.

5.4. Further Discussions

Sectoral Decomposition Is the aggregate pattern driven by a few sectors or does it reflect a broad-based phenomenon? To answer this question, we decompose clustered origins across twelve major industry groups. Figure 6 reveals substantial heterogeneity across sectors (see Table A1 for the classification). While clustered origins are generally positive at the aggregate level, individual sectors display markedly different dynamics. For example, the Information sector contributes little to the evolution of the U.S. economy aggregate volatility in recent decades.

Despite this heterogeneity, the major shifts in clustered origins are broadly shared across sectors. During the Great Moderation, comovement declined in most industries, including Finance & Insurance, Real Estate, Rental & Leasing, and Mining & Agriculture. This decline was followed by a widespread resurgence after 2000, particularly in Real Estate, Wholesale & Retail Trade, and Arts, Entertainment & Recreation. These results suggest that the evolution of clustered origins reflects economy-wide changes in firm synchronization rather than developments concentrated in a small number of sectors.

Labor Productivity We extend the analysis to labor productivity to assess whether clustered origins reflect fundamental firm performance beyond sales performance. Labor productivity-based estimates closely replicate the baseline sales results: the contribution of firm-specific productivity comovement declines during the Great Moderation and rises during the recessions of the 2000s (Panel a, Figure 7), and its share of aggregate variance increases from roughly 10% in the late 1990s to more than 25% during the Great Recession and subsequent recovery (Panel b). Although smaller in magnitude, these estimates confirm that clustered origins reflect broader firm-level synchronization rather than sales-specific dynamics.

Aggregate Fluctuations from Firm Comovement

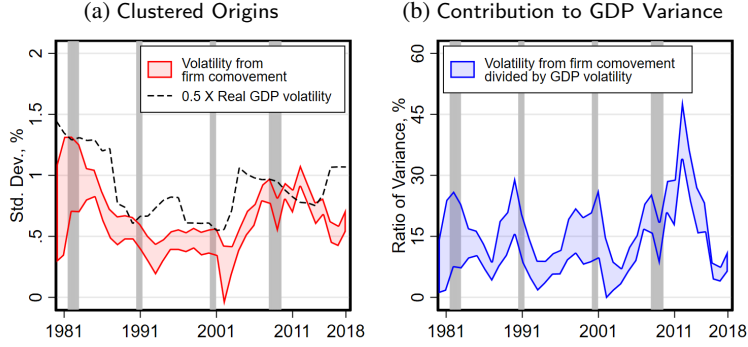


Figure 7: Clustered Origins with Labor Productivity Notes: This figure replicates the main analysis from Figure 3 using firm-level labor productivity instead of real sales. Panel (a) plots volatility levels, and Panel (b) plots the share of aggregate variance explained by the clustered origins of labor productivity.

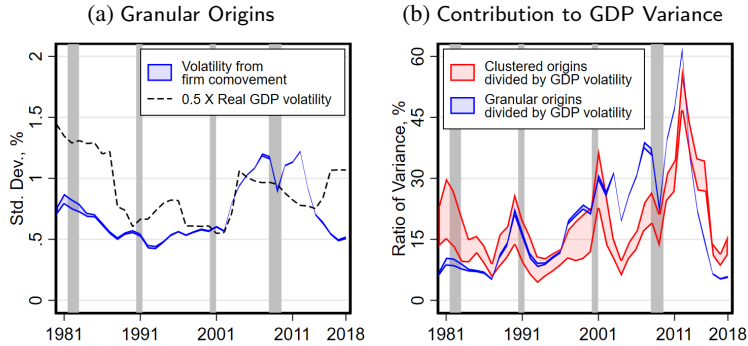


Figure 8: Granular Origins in U.S. Business Cycles Notes: The figure quantifies the contribution of volatile idiosyncratic fluctuations (the 'granular origin,' Γ_t) to aggregate U.S. business cycle volatility. Panel (a) plots the standard deviation of aggregate real GDP growth (black dashed line) against the contribution from granular origins (blue area). In Panel (b), the blue and red areas plot granular and clustered origins' contribution as a percentage of total GDP variance ($100 \times \Gamma_t / \text{var}(\widehat{\text{GDP}}_t)$ and $100 \times \chi_t / \text{var}(\widehat{\text{GDP}}_t)$), respectively. Shaded gray areas denote NBER recession dates.

Granular Origins Finally, we compare clustered origins with the granular origins, which attributes aggregate fluctuations to idiosyncratic shocks of large firms (e.g., Jovanovic, 1987; Gabaix, 2011; Carvalho and Gabaix, 2013, among many others). Figure 8 shows both similarities and differences between granular (blue area) and clustered origins (red area). Both channels contributed 10–15% of GDP variance through the 1980s and early 1990s, with their influence increasing after the mid-1990s. Our granular measure closely replicates prior estimates: the correction term $-h_t^2 \sigma_{A,t}^2$ in equation (22) is quantitatively negligible given the small squared HHI.¹⁶ The two channels nevertheless differ in their cyclical behavior: clustered origins peak consistently around recessions, while granular origins display different dynamics not exclusively tied to downturns. These distinct patterns suggest that the two channels respond to different types of economic shocks, pointing to the need for models that incorporate both idiosyncratic firm risk and correlated industry-level dynamics.

6. Conclusion

This paper addresses a central question in macroeconomics: what are the micro-level origins of the business cycle? We argue that the prevailing focus on aggregate shocks and the idiosyncratic shocks of granular firms overlooks a

¹⁶The difference between our measure and simpler demeaned approaches is proportional to $-h_t^2 \sigma_{A,t}^2$; because squared HHI is small, the adjustment is minor.

critical channel—the within-industry comovement of firm-specific components. Standard empirical practices, such as cross-sectional demeaning, are ill-suited to detect this mechanism, as they mechanically strip these correlations from the data.

Our contribution is to develop a novel, nonparametric bounds method that quantifies the importance of these clustered origins without restrictive assumptions. Applying this framework to U.S. publicly traded firms, we show that clustered origins are a significant and dynamic driver of aggregate fluctuations. They account for 10–20% of volatility in normal periods, rise sharply during downturns, and reach nearly 45% during the Great Recession. This channel provides a new microfounded perspective on major macroeconomic episodes, including the Great Moderation and the subsequent resurgence of volatility.

These findings reshape our view of the business cycle. Aggregate volatility is not solely driven by common shocks or the fortunes of a few dominant firms, but also by the synchronized fluctuations of many firms. While our approach establishes the quantitative importance of clustered origins, it is descriptive rather than causal and remains agnostic about underlying drivers. Multiple mechanisms are likely at play. Firm networks and input-output linkages (Oberfield, 2018; Bernard, Moxnes and Saito, 2019; Giroud and Mueller, 2019; Heise, 2024) can transmit shocks in a correlated manner. Earlier studies of industry-level interactions (Long and Plosser, 1983; Horvath, 1998; Horvath and Verbrugge, 1999; Dupor, 1999; Foerster et al., 2011; Atalay, 2017) highlight channels that may operate at the firm level. Other forces, including technology spillovers, vintage capital dynamics, and adoption effects (Bloom, Schankerman and Van Reenen, 2013; Mullen, 2020; Fiori and Scoccianti, 2021), may also contribute to firm comovements within industries. Disentangling the relative importance of these mechanisms remains an open and important question. Such work will help build a more complete microfounded theory of aggregate fluctuations and inform policy strategies aimed at enhancing macroeconomic stability.

A. Additional Simulation Exercise

First, we generate 5,000 firms' benchmark firm-specific components ($\varepsilon_{F,it}$) during 50 periods which are randomly generated from a multi-normal distribution with mean zero, 12% standard deviation, and 2.5% correlation. Figure A1 reports results with a time-invariant size distribution with fat-tails where the distribution is generated from Pareto distribution with shape parameter 1.2 on support $[1, \infty)$, which yields approximately a 13% Herfindahl-Hirschman index.

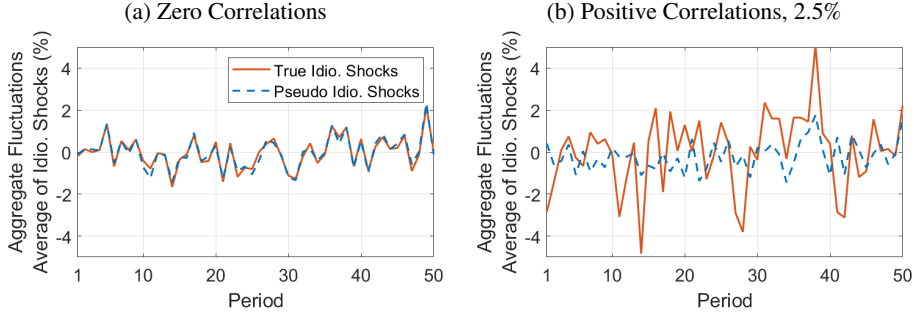


Figure A1: Benchmark vs. Pseudo firm-specific Components and Their Aggregate Fluctuations with fat tails Notes: The figure plots the aggregate fluctuations based on the average true and pseudo firm-specific components in each period (orange solid line, $\sum_{i=1}^{5,000} w_{it} \varepsilon_{F,it}$, and blue dashed line, $\sum_{i=1}^{5,000} w_{it} e_{F,it}$, respectively). In Panel (a), samples are generated from a multivariate normal distribution with zero mean, a 12% standard deviation, and a 2.5% pairwise correlation. In Panel (b), the pairwise correlation is set to zero.

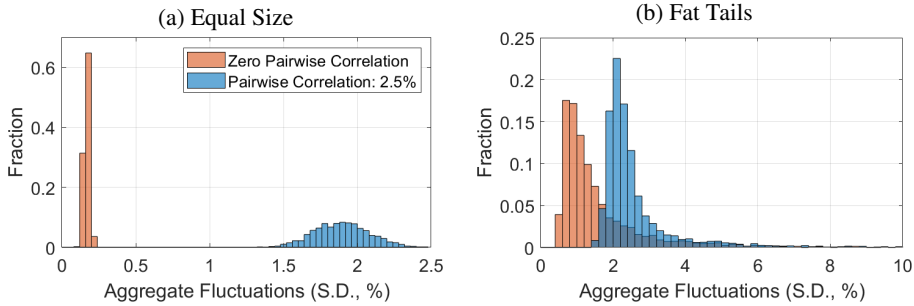


Figure A2: Sample Standard Deviations of Aggregate Fluctuations Notes: The figure plots histograms for the standard deviations of the aggregate fluctuations from 3,000 simulations. For the orange bars, samples are generated from a multivariate normal distribution with zero mean, a 12% standard deviation, and a 2.5% pairwise correlation. For the blue bars, the pairwise correlation is set to zero. Panel (a) assumes fixed and equal shares, $w_{it} = 1/N_t$, while Panel (b) draws firm shares from a Pareto distribution with shape parameter 1.2.

Compared to Figure 1, Figure A1 shows larger aggregate fluctuations due to granularity. Correlations still generate additional aggregate fluctuations. The solid orange line of Panel B with correlations is more volatile than Panel A's solid orange line without correlations. However, the dashed blue lines constructed by pseudo components have indistinguishable volatilities between Panels A and B. Second, we redo the above exercise 3,000 times and calculate the sample statistics of benchmark and pseudo firm-specific components. Figure A2 plots histograms for the aggregate volatility with and without unequal size distributions. Positive correlations lead to sizable aggregate fluctuations.

B. Arbitrary Weights in Section 3.4

Consider an arbitrary weight, $\{w_{it}\}_i$, satisfying $\sum_{i'} w_{i't} = 1$ and $w_{it} \geq 0$. Define pseudo common and firm-specific components based on the weighted mean, as follows.

$$e_{A,t}^w = \sum_{i'} w_{i't} \hat{y}_{i't} = \varepsilon_{A,t} + \sum_{i'} w_{i't} \varepsilon_{F,i't} \quad (\text{A1})$$

$$e_{F,it}^w = \hat{y}_{it} - e_{A,t} = \varepsilon_{F,it} - \sum_{i'} w_{i't} \varepsilon_{F,i't} \quad (\text{A2})$$

Then, the variance of pseudo firm-specific components is

$$\text{var}(e_{F,it}^w) = (1 - 2w_{it} + m_2^w)(1 - \rho_{F,t})\sigma_{F,t}^2, \quad (\text{A3})$$

where $m_2^w = \sum_{i'} w_{i't}^2 \in [N_t^{-1}, 1]$ measures how much equally weighted. Also, the pairwise correlation of pseudo firm-specific components are

$$\text{corr}(e_{F,it}^w, e_{F,i't}^w) = -\frac{w_{it} + w_{i't} - m_2^w}{\sqrt{1 - 2w_{it} + m_2^w} \sqrt{1 - 2w_{i't} + m_2^w}}, \quad (\text{A4})$$

which is independent of the pairwise correlation of benchmark firm-specific components denoted by $\rho_{F,t}$. If the weights are unequal, the firm-specific components correlate with the pseudo common in contrast to the equal weights case with homogeneous variance and covariance.

$$\text{corr}(e_{A,t}^w, e_{F,it}^w) = -\frac{w_{it} - m_2^w}{\sqrt{\frac{\sigma_{A,t}^2/\sigma_{F,t}^2 + \rho_{F,t}}{1 - \rho_{F,t}}} + m_2^w} \sqrt{1 - 2w_{it} + m_2^w}, \quad (\text{A5})$$

where the equal weights — $\forall i, w_{it} = N_t^{-1}$ — lead the pseudo common and firm-specific components to be uncorrelated for all firms; $\forall i, \text{corr}(e_{A,t}^w, e_{F,it}^w) = 0$.

C. Data and Measurements

[Step 1] We correct the industry-level deflators (p_{st}) — Chain-Type Price Indexes for Gross Output by Industry [2012=100] — from the U.S. Bureau of Economic Analysis (BEA) database. Sales (sale_{it}) and employments (emp_{it}) are directly from the Compustat North America: Fundamental Annuals (1975–2018) databases.

[Step 2] We construct the sample as follows. First, we keep the following observations in the Compustat database.

- No major mergers flag: Comparability status (compst_{it}) does not equal to AB .
- Country ISO 3 digit code (loc_{it}): USA
- Currency ISO 3 digit code (curcd_{it}): USD

Then, we exclude firms with the following criteria.

- Non-positive sales
- Non-positive employments
- Utilities sector (NAICS 22)
- Public administration sector (NAICS 91–92)

[Step 3] We merge the Compustat sample and the industry-level BEA deflator using Table A2. We calculate the logged labor productivity as real sales divided by employments ($\ln \text{sale}_{it} - \ln p_{st} - \ln \text{emp}_{it}$) for firm i in industry s at t .

[Step 4] Since some clusters have low observations, we merge them. See Table A2 for the list of clusters.

D. Details on Factor Model Interpretation for Firm-Specific Components

This section explores an alternative approach to address pairwise correlations in the framework discussed in the preceding sections by introducing a factor model with heterogeneous factor loadings. In this framework, comovements among firm-specific components are driven by heterogeneous reactions to a latent common factor. We show that, under the standard factor-loading normalization, the estimated factor model faces the same problem as the demeaned process discussed earlier: it mechanically removes firm comovements and their contribution to aggregate volatility.

Framework We begin by decomposing the benchmark firm-specific component into the latent factor and pairwise uncorrelated components, i.e., $\varepsilon_{F,it} = \lambda_{it}f_t + u_{it}$ where $\varepsilon_{A,t}$, f_t , u_{it} and $u_{i't}$ are uncorrelated for any firm $i \neq i'$. In this factor model, it is expressed as:

$$\hat{y}_{it} = \varepsilon_{A,t} + \underbrace{\lambda_{it}f_t + u_{it}}_{=\varepsilon_{F,it}}, \quad (\text{A6})$$

where heterogeneous factor loadings imply that pairwise correlations between $\varepsilon_{F,it}$ and $\varepsilon_{F,i't}$ vary across firms.

This representation provides an alternative formulation of the business-cycle components. The corresponding variance and covariance are:

$$\text{var}(\hat{y}_{it}) = \sigma_{A,t}^2 + \text{var}(\lambda_{it}f_t + u_{it}) = \sigma_{A,t}^2 + \lambda_{it}^2\sigma_{f,t}^2 + \sigma_{u,it}^2, \quad (\text{A7})$$

$$\text{cov}(\hat{y}_{it}, \hat{y}_{i't}) = \sigma_{A,t}^2 + \text{cov}(\lambda_{it}f_t, \lambda_{i't}f_t) = \sigma_{A,t}^2 + \lambda_{it}\lambda_{i't}\sigma_{f,t}^2, \quad (\text{A8})$$

where $\lambda_{it}^2\sigma_{f,t}^2 + \sigma_{u,it}^2$ and $\lambda_{it}\lambda_{i't}\sigma_{f,t}^2$ are the corresponding part of $\sigma_{F,it}^2$ and $\rho_{FF,iit'}\sigma_{F,it}\sigma_{F,i't}$ previously seen in equations (19) and (20) within our benchmark framework.

Pseudo Common and Firm-Specific Components Identification in latent factor models requires a normalization on factor loadings, typically imposing a zero cross-sectional mean (see Bai and Ng, 2013). Under this normalization, equation (7) implies the following decomposition:

$$\hat{y}_{it} = (\varepsilon_{A,t} + \bar{\lambda}_t f_t) + \underbrace{(\lambda_{it} - \bar{\lambda}_t) f_t}_{\tilde{\lambda}_{i,t}} + u_{it} \quad (\text{A9})$$

$$= (\bar{y}_t) + [\hat{y}_{it} - \bar{y}_t] \equiv e_{A,t} + e_{F,it}, \quad (\text{A10})$$

where $\bar{\lambda}_t = N_t^{-1} \sum_i \lambda_{it}$ and $0 = N_t^{-1} \sum_i \tilde{\lambda}_{i,t}$. The conventional factor model thus estimates $(\tilde{\lambda}_{i,t} \equiv \lambda_{it} - \bar{\lambda}_t)$ rather than λ_{it} , linking the estimated firm-specific component to $e_{F,it}$.

The discrepancy between the benchmark and pseudo components does not generally vanish asymptotically. Specifically,

$$\mathbb{E}[|e_{A,t} - \varepsilon_{A,t}|^2] = \mathbb{E}[|e_{F,it} - \varepsilon_{F,it}|^2] = \frac{1}{N_t} \bar{\sigma}_{u,t}^2 + \bar{\lambda}_t^2 \sigma_{f,t}^2, \quad (\text{A11})$$

The pseudo components also fail to consistently estimate the variances of benchmark components:

$$\text{var}(e_{A,t}) = \sigma_{A,t}^2 + \bar{\lambda}_t^2 \sigma_{f,t}^2 + \frac{1}{N_t} \bar{\sigma}_{u,t}^2 \quad (\text{A12})$$

$$\text{var}(e_{F,it}) = \left(1 - \frac{1}{N_t}\right) \sigma_{u,it}^2 + \tilde{\lambda}_{i,t}^2 \sigma_{f,t}^2 - \frac{1}{N_t} (\sigma_{u,it}^2 - \bar{\sigma}_{u,t}^2), \quad (\text{A13})$$

where $\bar{\sigma}_{u,t}^2 = N_t^{-1} \sum_i \sigma_{u,it}^2$ is the average variance. These expressions generally do not recover the true common and idiosyncratic variances ($\text{var}(\varepsilon_{A,t})$ and $\text{var}(\varepsilon_{F,it})$) unless true pairwise comovements of benchmark components are nearly absent (i.e., when $\bar{\lambda}_t = 0$).

Firm Comovements Finally, it's worth noting that the pairwise covariance of pseudo firm-specific components tends to disregard the actual covariance:

$$\text{cov}(e_{F,it}, e_{F,i't}) = \tilde{\lambda}_{i,t} \tilde{\lambda}_{i',t} \sigma_{f,t}^2 - \frac{1}{N_t} \bar{\sigma}_{u,t}^2 - \frac{1}{N_t} \left[(\sigma_{u,it}^2 - \bar{\sigma}_{u,t}^2) + (\sigma_{u,i't}^2 - \bar{\sigma}_{u,t}^2) \right], \quad (\text{A14})$$

which cross-sectional average converges to zero as $N_t \rightarrow \infty$, implying asymptotically uncorrelated pseudo firm-specific components regardless of the true underlying comovement of benchmark components.

The conclusions from the previous sections therefore continue to hold when pairwise comovements arise from heterogeneous exposure to a common latent factor. Researchers should be cautious when using sample demeaning or zero-normalization restrictions (as in Bai and Ng, 2013), as these procedures can mechanically eliminate firm interdependencies and understate their contribution to aggregate volatility.

E. Additional Figures and Tables

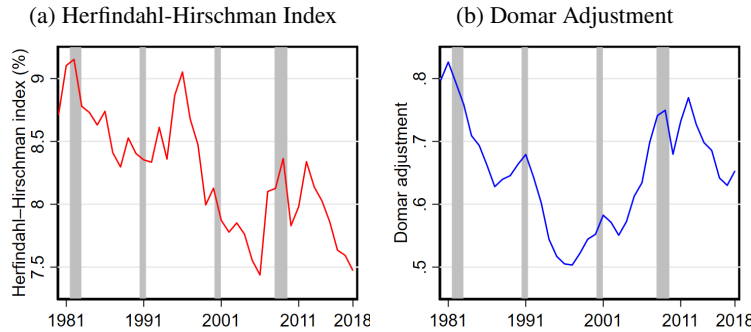


Figure A3: Herfindahl-Hirschman Index and Domar Adjustment

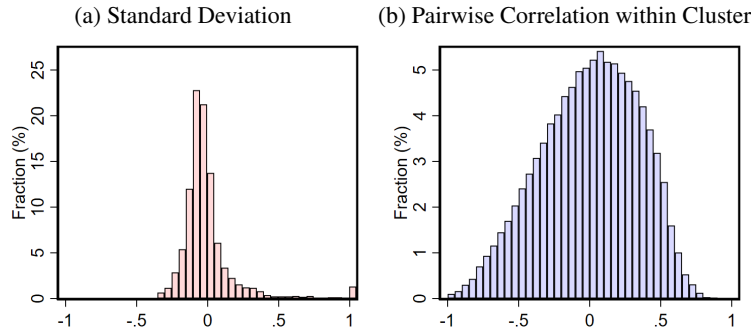


Figure A4: Histograms: Firm Volatility and Comovements (Demeaned Across Clusters) Notes: The figures plot histograms of within-cluster demeaned firm volatility (Panel a) and pairwise correlations (Panel b) for 1995. Moments are first calculated for each firm (standard deviation) and each intra-cluster firm pair (correlation) using data from the 1990–2000 window. We then demean these values by subtracting the corresponding cluster-specific mean. The plots visualize the remaining cross-sectional variation.

Aggregate Fluctuations from Firm Comovement

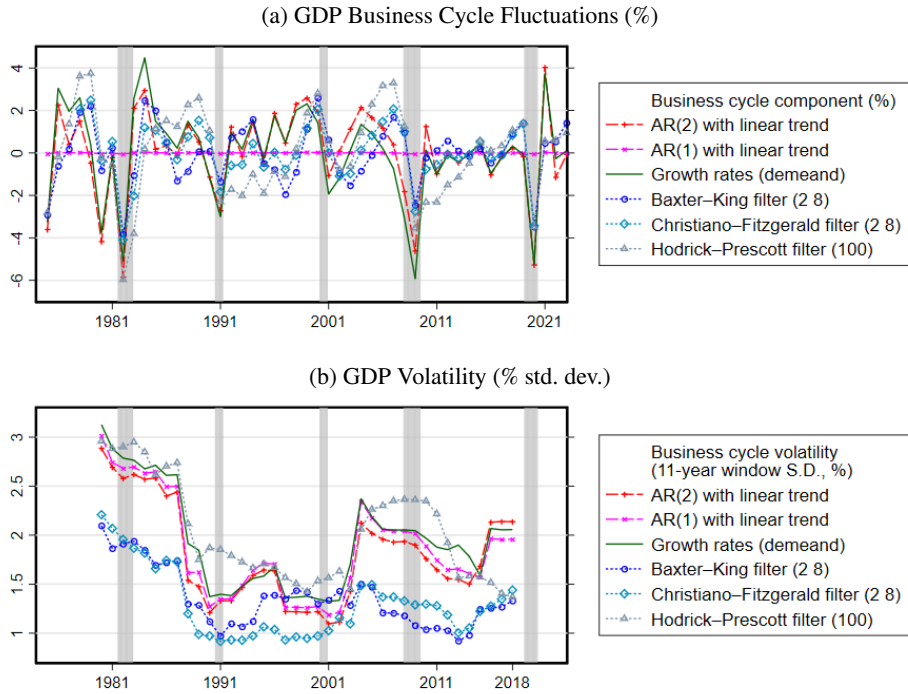


Figure A5: U.S. Business Cycle Fluctuations Notes: The figure plots the aggregate U.S. business-cycle measurements and their volatility under various specifications. Panel (a) plots the business-cycle component of annual U.S. private GDP. Panel (b) reports the standard deviation computed over an 11-year rolling window.

Table A1
List of Sectors

Sector	Clusters
Mining, agriculture, forestry, fishing, & hunting	1–5
Manufacturing (Durable)	6–15
Manufacturing (Nondurable)	16–23
Wholesale & retail trade	24–25
Transportation & warehousing	26–31
Information	32–35
Finance & insurance	36–39
Real estate, rental & leasing	40–41
Professional & business services	42–45
Educational services, health care & social assistance	46–48
Arts, entertainment, recreation, accommodation & food services	49–51
Other services, except government	52

Aggregate Fluctuations from Firm Comovement

Table A2
List of Clusters

Cluster	BEA Industry	NAICS	
		1997	2017
1. Agriculture, forestry, fishing, and hunting	· Farms · Forestry, fishing, and related activities	111–2 113–5	111–2 113–5
2. Oil and gas extraction	· Oil and gas extraction	211	211
3. Mining, except oil and gas	· Mining, except oil and gas	212	212
4. Support activities for mining	· Support activities for mining	213	213
5. Construction	· Construction	230	230
6. Wood products	· Wood products	321	321
7. Nonmetallic mineral products	· Nonmetallic mineral products	327	327
8. Primary metals	· Primary metals	331	331
9. Fabricated metal products	· Fabricated metal products	332	332
10. Machinery	· Machinery	333	333
11. Computer and electronic products	· Computer and electronic products	334	334
12. Electrical equipment, appliances, and components	· Electrical equipment, appliances, and components	335	335
13. Motor vehicles, bodies and trailers, and parts, and Other transportation	· Motor vehicles, bodies and trailers, and parts · Other transportation equipment	3361–6 3369	3361–6 3369
14. Furniture and related products	Furniture and related products	337	337
15. Miscellaneous manufacturing	Miscellaneous manufacturing	339	339
16. Food and beverage and tobacco products	· Food and beverage and tobacco products	311–2	311–2
17. Textile mills and textile product mills	· Textile mills and textile product mills	313–4	313–4
18. Apparel and leather and allied products	· Apparel and leather and allied products	315–6	315–6
19. Paper products	· Paper products	322	322
20. Printing and related support activities	· Printing and related support activities	323	323
21. Petroleum and coal products	· Petroleum and coal products	324	324
22. Chemical products	· Chemical products	325	325
23. Plastics and rubber products	· Plastics and rubber products	326	326
24. Wholesale trade	· Wholesale trade	420	420
25. Retail trade	· Motor vehicle and parts dealers · Food and beverage stores · General merchandise stores · Other retail	441 445 452 442–4, 446–8, 451, 453–4	441 445 452 442–4, 446–8, 451, 453–4
26. Air transportation	· Air transportation	481	481
27. Rail transportation	· Rail transportation	482	482
28. Water transportation	· Water transportation	483	483

Continued on next page

Aggregate Fluctuations from Firm Comovement

Table A2 — continued from previous page

Cluster	BEA Industry	NAICS	
		1997	2017
29. Truck transportation	· Truck transportation	484	484
30. Pipeline transportation	· Pipeline transportation	486	486
31. Other transportation (transit and ground) and support activities, and Warehousing and storage	· Transit and ground passenger transportation	485	485
	· Other transportation and support activities	487-8, 491-2	487-8, 491-2
	· Warehousing and storage	493	493
32. Publishing industries, except internet (includes software)	· Publishing industries, except internet (includes software)	511	511
33. Motion picture and sound recording industries	· Motion picture and sound recording industries	512	512
34. Broadcasting and telecommunications	· Broadcasting and telecommunications	513	515, 517
35. Data processing, internet publishing, and other information services	Data processing, internet publishing, and other information services	514	518-9
36. Federal Reserve banks, credit intermediation, and related activities	· Federal Reserve banks, credit intermediation, and related activities	521-2	521-2
37. Securities, commodity contracts, and investments	· Securities, commodity contracts, and investments	523	523
38. Insurance carriers and related activities	· Insurance carriers and related activities	524	524
39. Funds, trusts, and other financial vehicles	· Funds, trusts, and other financial vehicles	525	525
40. Real estate	· Real estate	531	531
41. Rental and leasing services and lessors of intangible assets	· Rental and leasing services and lessors of intangible assets	532-3	532-3
42. Computer systems design and related services	· Computer systems design and related services	5415	5415
43. Legal services, and miscellaneous professional, scientific, and technical services	· Legal services	5411	5411
	· Miscellaneous professional, scientific, and technical services	5412-4, 5416-9	5412-4, 5416-9
44. Administrative and support services	· Administrative and support services	561	561
45. Waste management and remediation services	· Waste management and remediation services	562	562
46. Educational services	· Educational services	610	610
47. Ambulatory health care services	· Ambulatory health care services	621	621
48. Hospitals, Nursing and residential care facilities, and social assistance	· Hospitals	622	622
	· Nursing and residential care facilities	623	623
	· Social assistance	624	624
49. Arts, entertainment, and recreation	· Performing arts, spectator sports, museums, and related activities	711-2	711-2
	· Amusements, gambling, and recreation industries	713	713
50. Accommodation	· Accommodation	721	721
51. Food services and drinking places	· Food services and drinking places	722	722
52. Other services, except government	· Other services, except government	810	810

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